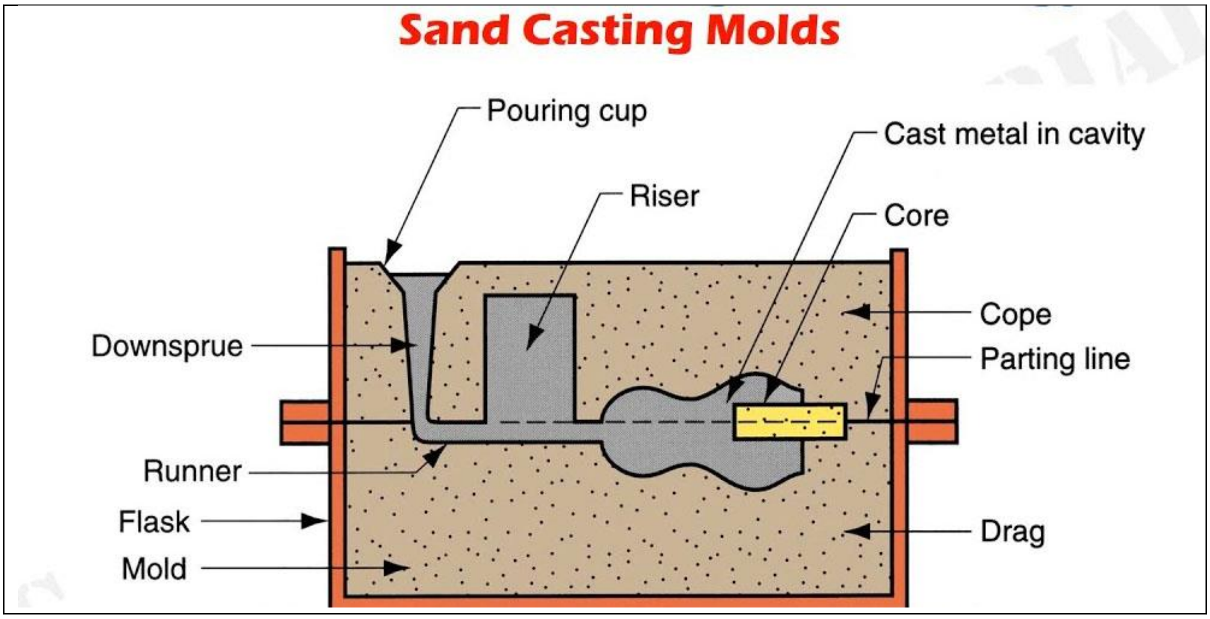
Unit-II

Manufacturing Processes: Principles of Casting, Forming, joining processes, Machining, Introduction to CNC machines, 3Dprinting, and Smart manufacturing.

Thermal Engineering – Working principle of Boilers, Otto cycle, Diesel cycle, Refrigeration and air – conditioning cycles, IC engines, 2-Stroke and 4-Stroke engines, SI/CI Engines, Components of Electric and Hybrid Vehicles.

Casting process:

Casting is a manufacturing process in which a molten liquid material is usually poured into a mould, which contains a hollow cavity of the desired shape, and then allowed to solidify. The solidified part is also known as a casting, which is ejected or broken out of the mould to complete the process.



Types of casting process:

- Sand casting
- Gravity casting
- Pressure Die Casting
- Investment Casting
- Plaster Casting
- Centrifugal Casting
- Lost-Foam Casting
- Vacuum Casting

Applications:

- Transport: Automobile, aerospace, railways and shipping.
- Heavy Equipment: Construction, farming and mining.
- Machine Tools: Machining, casting, plastics moulding, forging, extrusion and forming.
- Plant Machinery: Chemical, petroleum, paper, sugar, textile, steel and thermal plants.

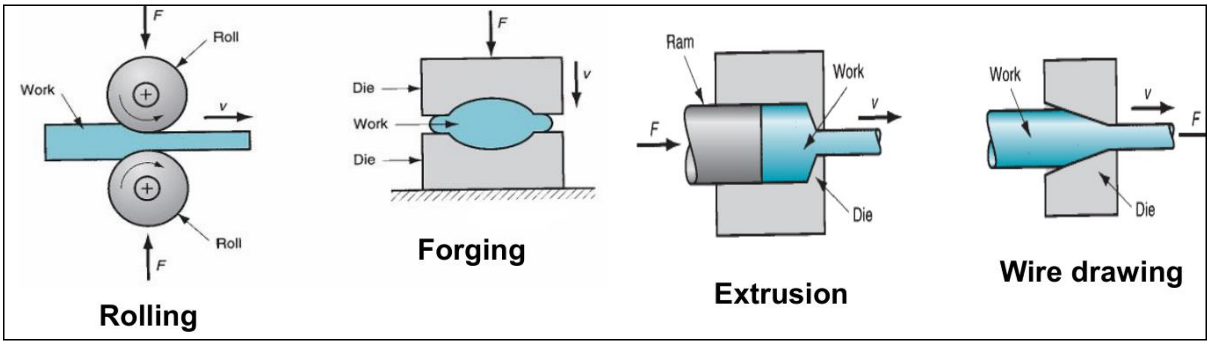
Forming:

Metal forming is a process where materials are subjected to plastic deformation to obtain the required size, shape, and change the physical and chemical properties.

Main types of forming processes:

1. Rolling process 2. Forging process

3. extrusion process 4. Drawing process



Rolling process: Rolling is a metal forming process in which metal stock is passed through one or more pairs of rolls to reduce the thickness, to make the thickness uniform, and/or to impart a desired mechanical property.

Forging process: Forging is a manufacturing process involving the shaping of a metal through hammering, pressing, or rolling. These compressive forces are delivered with a hammer or die.

Extrusion process: In extrusion process, metal or work piece is forced (pushing) to flow through a die to reduce its cross section or convert it into desire shape. This process is extensively used in pipes and steel rods manufacturing.

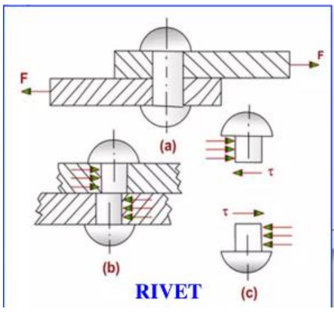
Drawing process: Drawing is a manufacturing process that uses tensile forces to elongate metal, glass, or plastic. As the material is drawn (pulled), it stretches and becomes thinner, achieving a desired shape and thickness.

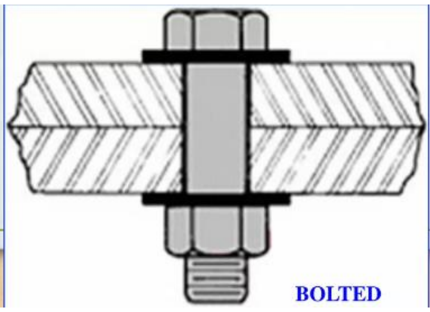
Joining process:

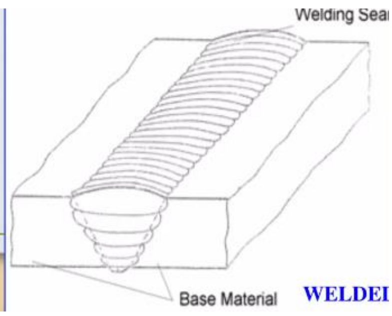
Joining processes are characterized by their ability to fuse or "join" two or more components for the purpose of creating a different material with help of heat, pressure, rivets, adhesive and bolts &nuts(screws).

Types of joining process:

- Welding (permanent joining)
- Soldering (permanent joining)
- Brazing /braze welding (permanent joining)
- Riveting (permanent joining)
- Adhesive bonding (semi-permanent joining)
- Bolts and nuts/screws (temporary/semi-permanent joining)







Welding: Welding is a process of permanently joining materials. Welding joins different materials alloys by number of processes, in which heat is supplied either electrically or by means of torch. Welding is done by application of heat or both heat and pressure.

Soldering: soldering is a metal joining process in which the filler metal or alloy is heated to a temperature below 450°C and melted. Only filler metal melts fusing the work piece. Solder is an alloy of tin (63%) and lead (37%).

Brazing: Brazing is a metal joining process in which the filler metal or alloy is heated to a temperature above 450°C and melted. Only filler metal melts fusing the work piece.

Riveting: In this metal joining process, the metallic parts are joined together by use of rivets. In this process, the metallic parts to be joined do not undergo any change in physical or atomic structure.

Adhesive bonding: Adhesive bonding is a process in which joining between two or more parts is accomplished by the solidification or hardening of a non-metallic adhesive material, placed between the faying surfaces of the parts.

Bolting: In this metal joining process, the metallic parts are joined together by bolt/screw (and/or) nut. This process is widely used in assembling of parts are to be joined temporarily or joints which require periodic maintenance.

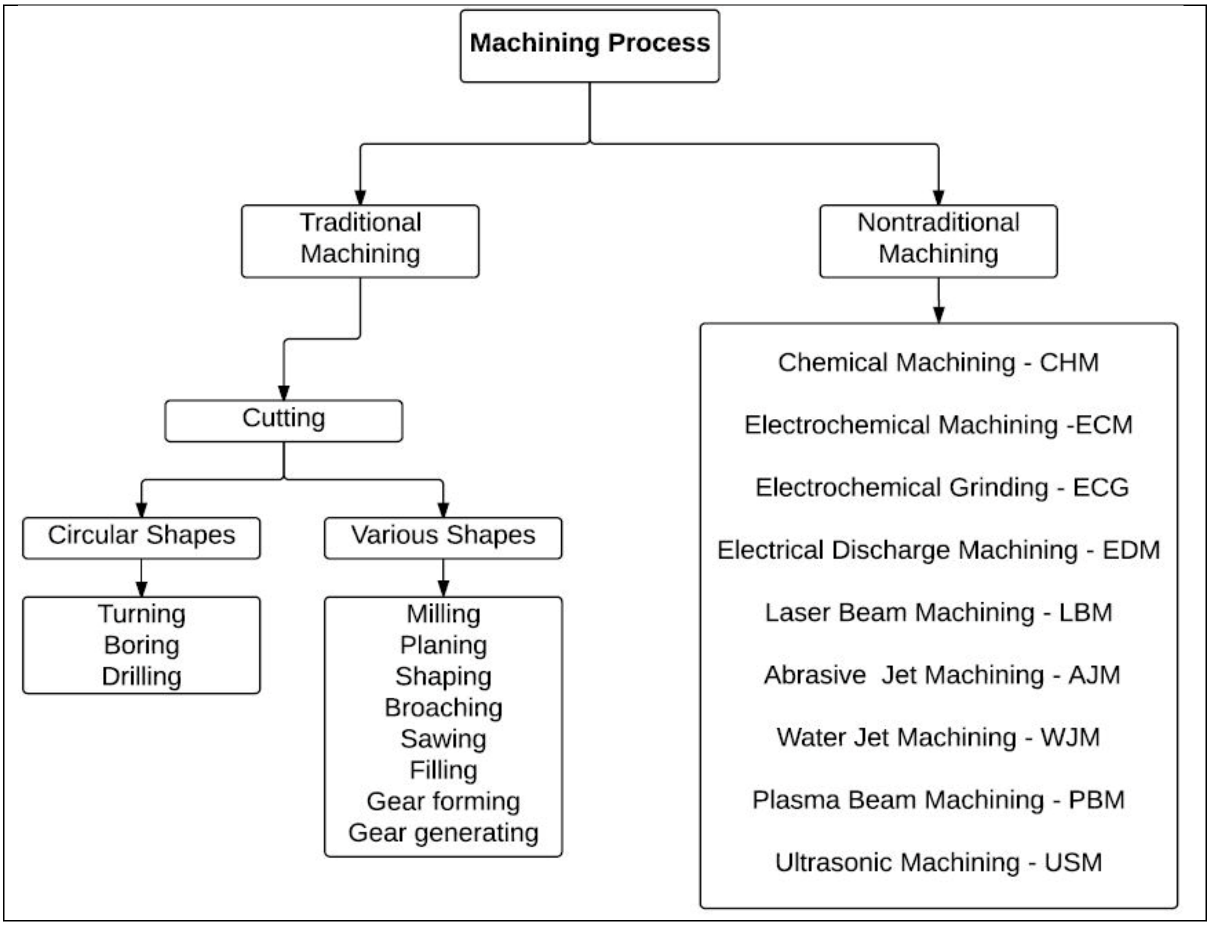
Machining:

Machining is a process in which a material is cut to a desired final shape and size by a controlled material-removal process.

Machining is a subtractive manufacturing process that involves the use of cutting tools, discs, abrasive wheels, and more to remove excess material from a workpiece. Furthermore, this process is used to remove unwanted materials to achieve the desired product shape.

Types of machining process:

Traditional machining process: this process involves the use of cutting tools to remove excess materials from a workpiece on direct contact. Furthermore, this machining operation occurs when the cutting tool directly interacts with the workpiece.



Non-traditional machining process: the machines used in this process do not require direct contact with the cutting material. This type of precision machining process does not require contact with the workpiece to remove material from it.

CNC machining:

Computer Numerical Control (CNC) machining is a manufacturing process in which pre-programmed computer software dictates the movement of factory tools and machinery.

Basic steps in CNC machining process:

There are 4 main basic steps:

- ✓ Step 1: Prepare a CAD Model. This involves creating a 2D or 3D model design of the product.
- ✓ Step 2: Conversion to CNC Compatible Format. CNC machines won't understand the CAD model.
- ✓ Step 3: Setup Execution.
- ✓ Step 4: Machine the Part.

Types of machining operations using CNC machine:

- ➢ CNC Lathes machine
- ➢ CNC Turning machine
- ➢ CNC Drilling machine
- ➢ CNC Milling machine
- ➢ CNC plasma cutters
- ➢ CNC Laser cutting machines

Applications:

- ❖ Medical, aerospace, industrial, oil and gas, hydraulics, firearms etc.

3D printing:

3D printing, also known as additive manufacturing, is a method of creating a three-dimensional object layer-by-layer using a computer created design. 3D printing is an additive process whereby layers of material are built up to create a 3D part.

3D printing technologies:

There are three broad types of 3D printing technology; **sintering**, **melting**, and **stereolithography**.

Sintering is a technology where the material is heated, but not to the point of melting, to create high resolution items. Metal powder is used for direct metal laser sintering while thermoplastic powders are used for selective laser sintering.

Melting methods of 3D printing include powder bed fusion, electron beam melting and direct energy deposition, these use lasers, electric arcs or electron beams to print objects by melting the materials together at high temperatures.

Stereolithography utilises photopolymerization to create parts. This technology uses the correct light source to interact with the material in a selective manner to cure and solidify a cross section of the object in thin layers.

Applications:

- Prosthetics (artificial device that replace a missing body part)
- Replacement Parts

- Implants
- Emergency Structures
- Aeronautics and Space Travel
- Pharmaceuticals
- Custom Clothing
- Custom-Fitted Personal Products

Smart manufacturing: Smart manufacturing is a broad category of manufacturing that employs computer-integrated manufacturing, high levels of adaptability and rapid design changes, digital information technology, and more flexible technical workforce training.

Essence of smart manufacturing: The essence of smart manufacturing is captured in six pillars:

1.Manufacturing technology and processes	2.Materials
3.Data	4.Predictive engineering
5.Sustainability and resource sharing	6.Networking

Material handling and supply chains have been an integral part of manufacturing.

Applications:

- ✓ *3D printing (additive manufacturing)*
- ✓ *Artificial intelligence*
- ✓ *Digital manufacturing*
- ✓ *Cyber manufacturing*

Basic heat engine, refrigeration and heat pump:

Heat engine: Converting heat (thermal energy) to work (mechanical energy) requires the use of some special devices. These devices are called heat engines.

1. They receive heat from a high-temperature source (solar energy, oil furnace, nuclear reactor, etc.).

2. They convert part of this heat to work (usually in the form of a rotating shaft).

3. They reject the remaining waste heat to a low-temperature sink (the atmosphere, rivers, etc.).

4. They operate on a cycle.

Efficiency: The fraction of the heat input that is converted to network output is a measure of the performance of a heat engine and is called the thermal efficiency.

Thermal efficiency = $\frac{\text{Net work output}}{\text{Total heat input}} = \frac{W_{net}}{Q_H}$ $(W_{net} = Q_H - Q_L)$

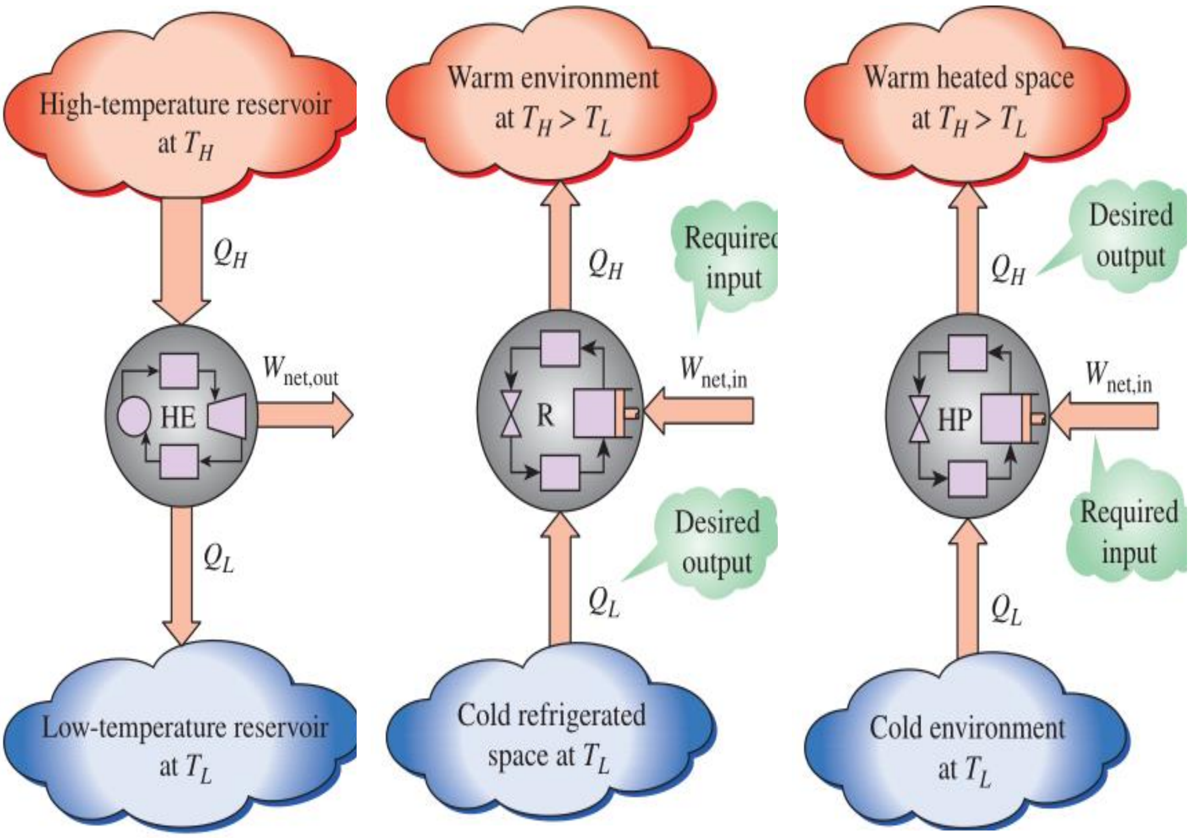
$\eta_{th} = \frac{Q_h - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$ or $\eta_{th} = \frac{T_h - T_L}{T_H} = 1 - \frac{T_L}{T_H}$ (Only for reversible cycles)

Refrigerator: The transfer of heat from a low-temperature medium to a high-temperature one requires special devices called refrigerators.

COP: The efficiency of a refrigerator is expressed in terms of the coefficient of performance (COP), denoted by $(COP)_R$. The objective of a refrigerator is to remove heat (Q_L) from the refrigerated space. To accomplish this objective, it requires a work input of $W_{net,in}$. Then the COP of a refrigerator can be expressed as

Thermal efficiency = $\frac{\text{Desired output}}{\text{Required work input}} = \frac{Q_L}{W_{net}}$ $(W_{net} = Q_H - Q_L)$

$COP_R = \frac{Q_L}{W_{net}} = \frac{Q_L}{Q_H - Q_L}$ or $COP_R = \frac{T_L}{T_H - T_L}$



Heat engine ***Refrigerator*** ***Heat pump***

Boiler: A boiler may be defined as a closed vessel in which steam is produced from water by combustion of fuel.

The steam generated is employed for the following purposes:

(i) For generating power in steam engines or steam turbines.

(ii) In the textile industries for sizing and bleaching etc., and many other industries like sugar mills; chemical industries.

(iii) For heating the buildings in cold weather and for producing hot water for hot water supply.

Working principle: The boiler is a closed vessel in which the water is stored. Hot gases are produced by burning fuel in the furnace. These hot gases are made to come in contact with the water vessel where the heat transfer takes place between the water and the steam. Therefore, the basic principle of the boiler is to convert water into steam by using heat energy. There are different types of boilers used for different purposes.

- In the fire tube boilers, the hot gases are inside the tubes and the water surrounds the tubes.
- In the water tube boilers, the water is inside the tubes and hot gases surround them.

Engine: It is a device which converts one form of energy in to another form.

Heat Engine: It is a device which derives heat energy from the combustion of fuel and converts this energy into mechanical work is termed as a heat engine.

Classification of Heat Engines:

• **Internal Combustion Engines (IC Engines)**

In IC engines, combustion of fuel takes place inside the engine cylinder.

Examples: Diesel Engines, Petrol Engines, Gas engines.

• **External Combustion Engines (EC Engines)**

In EC engines, combustion of fuel takes place outside the working cylinder.

Examples: Steam Engines and Steam turbines

Applications of I.C. engines

The I.C. engines are generally used for :

i) Road vehicles (e.g., scooter, motorcycle, buses etc.)

ii) Air craft iii) Locomotives

iv) Construction in civil engineering equipment such as bull-dozer, scraper, power shovels etc.

v) Pumping sets vi) Several industrial applications

different parts of I.C. engines

Parts common to both petrol and diesel engine:

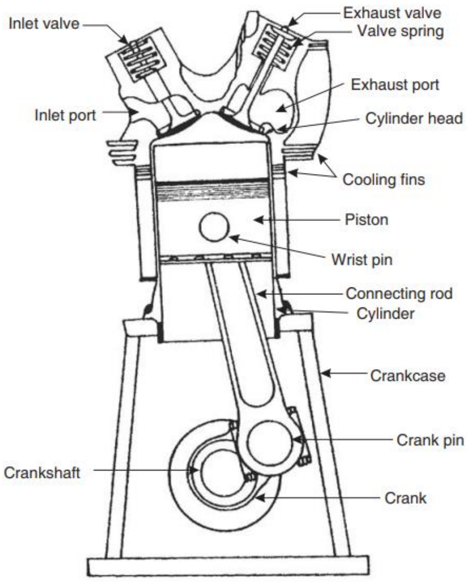
1.Cylinder
2. Cylinder head
3. Piston
4. Piston rings
5. Gudgeon pin
6. Connecting rod
7. Crank
8. Crankshaft
9. Engine bearing 1
10. Crankcase
11. Flywheel
12. Governor
13. Valves and valve operating mechanism.

Parts for petrol engines only:

1. Spark plugs
2. Carburetor
3. Fuel pump.

Parts for Diesel engine only:

1. Fuel pump.
2. Injector.



Terms connected with IC engines

Bore. The inside diameter of the cylinder is called bore.

Stroke. As the piston reciprocates inside the engine cylinder, it has got limiting upper and lower positions beyond which it cannot move and reversal of motion takes place at these limiting positions.

The linear distance along the cylinder axis between two limiting positions, is called stroke.

Top Dead Centre (T.D.C.). The top most position of the piston towards cover end side of the cylinder is called “top dead centre”.

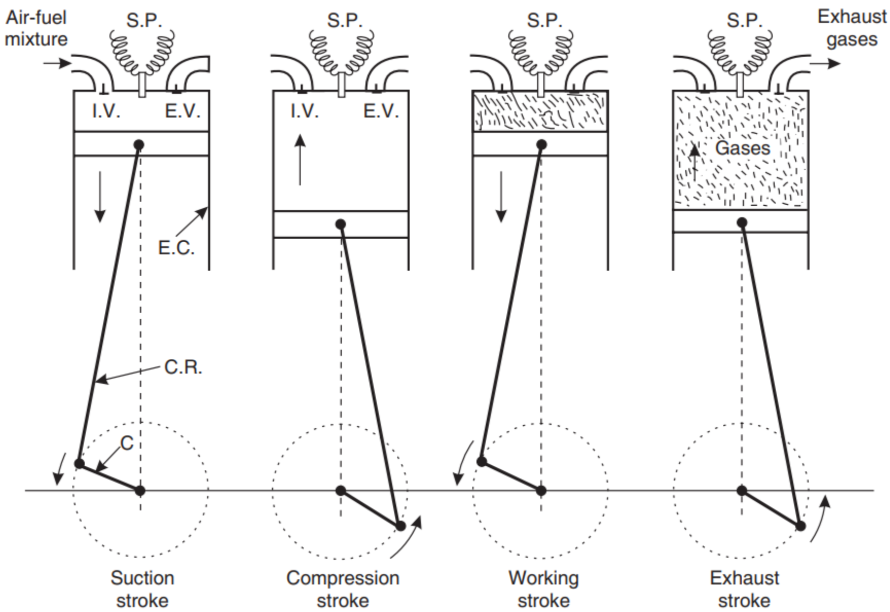
Bottom Dead Centre (B.D.C.). The lowest position of the piston towards the crank end side of the cylinder is called “bottom dead centre

Clearance volume. The volume contained in the cylinder above the top of the piston, when the piston is at top dead centre, is called the clearance volume.

Swept volume. The volume swept through by the piston in moving between top dead centre and bottom dead centre, is called swept volume or piston displacement. Thus, when piston is at bottom dead centre, total volume = swept volume + clearance volume.

Compression ratio. It is ratio of total cylinder volume to clearance volume

Working of Four-Stroke Spark Ignition (SI) Engines/ Petrol Engine:



I.V = Intel valve, E.V. = Exhaust valve, E.C. = Engine cylinder, C.R. = Connecting rod
C = Crank, S.P. = Spark plug.

(a) Suction Stroke (First Stroke of the Engine)

• Piston moves down from TDC to BDC

• Inlet valve is opened and the exhaust valve is closed.

• Pressure inside the cylinder is reduced below the atmospheric pressure.

• The mixture of air fuel is sucked into the cylinder through the inlet valve.

(b) Compression Stroke: (Second Stroke of the piston)

• Piston moves up from BDC to TDC

• Both inlet and exhaust valves are closed.

• The air fuel mixture in the cylinder is compressed.

(c)Working or Power or Expansion Stroke: (Third Stroke of the Engine)

• The burning gases expand rapidly. They exert an impulse (thrust or force) on the piston. The piston is pushed from TDC to BDC

• This movement of the piston is converted into rotary motion of the crankshaft through connecting rod.

• Both inlet and exhaust valves are closed.

(d) Exhaust Stroke (Fourth stroke of the piston)

• Piston moves upward from BDC

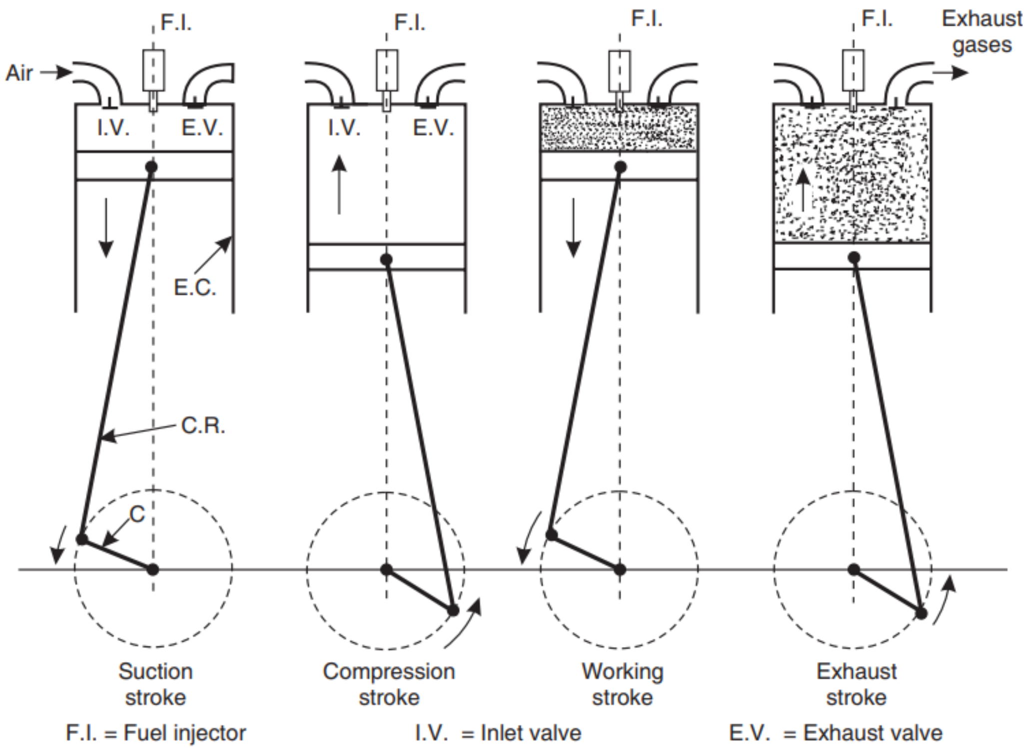
• Exhaust valve is opened and the inlet valve is closed.

• The burnt gases are forced out to the atmosphere through the exhaust valve (Some of the burnt gases stay in the clearance volume of the cylinder)

• The exhaust valve closes shortly after TDC

• The inlet valve opens slightly before TDC and the cylinder is ready to receive fresh charge to start a new cycle.

Working of Four-Stroke Compression Ignition (CI) Engines/ Diesel Engine:



(a) Suction Stroke (First Stroke of the piston)

• Piston moves from TDC to BDC

• Inlet valve is opened and the exhaust valve is closed.

• The pressure inside the cylinder is reduced below the atmospheric pressure.

• Fresh air from the atmosphere is sucked into the engine cylinder through air cleaner and inlet valve.

(b) Compression stroke (Second stroke of the piston)

• Piston moves from BDC to TDC

• Both inlet and exhaust valves are closed.

• The air is drawn during suction stroke is compressed to a high pressure and temperature

(c) Working or power or expansion stroke (Third stroke of the piston)

• The burning gases (products of combustion) expand rapidly.

• The burning gases push the piston move downward from TDC to BDC

• This movement of piston is converted into rotary motion of the crank shaft through connecting rod.

• Both inlet and exhaust valves are closed.

(d) Exhaust Stroke (Fourth stroke of the piston)

• Piston moves from BDC to TDC

• Exhaust valve is opened the inlet valve is closed.

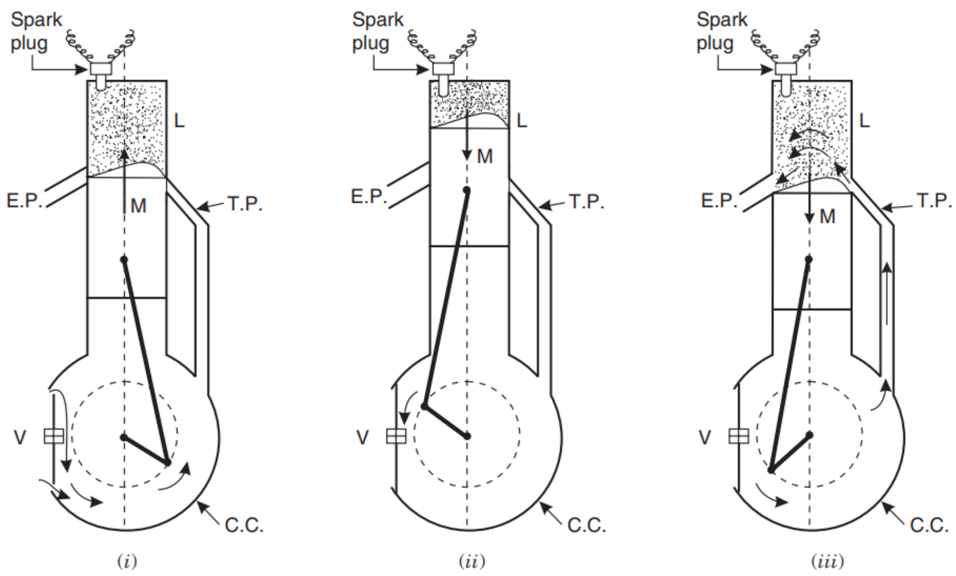
• The burnt gases are forced out to the atmosphere through the exhaust valve. (some of the burnt gases stay in the clearance volume of the cylinder)

• The exhaust valve closes shortly after TDC

• The inlet valve opens slightly before TDC and the cylinder is ready to receive fresh air to start a new cycle.

TWO-STROKE ENGINES

In this engine suction and exhaust strokes are eliminated. Here instead of valves, ports are used. The exhaust gases are driven out from engine cylinder by the fresh charge of fuel entering the cylinder nearly at the end of the working stroke.



L = Cylinder E.P. = Exhaust port T.P. = Transfer port V = valve, C.C = Crank chamber

First Stroke (Upward Stroke of the piston)

(a) Compression and inductance:

• The piston moves upwards from Bottom Dead Centre (BDC) to Top Dead Centre (TDC).

• Both transfer and exhaust ports are covered.

• Air which is transferred already into the engine cylinder is compressed by moving piston.

• The pressure and temperature of the air increases.

• At the same time, fresh air is admitted into the crankcase through the inlet port /plate valve

(b) Ignition and inductance.

• Piston almost reaches the top dead centre.

• The fuel is injected into the hot compressed air inside the cylinder. The fuel mixed with hot air and burns.

• The admission of fresh air into the crankcase continues till the piston reaches the top centre.

Second Stroke (Downward Stroke of the piston)

(c) Expansion and crank case compression:

• The burning gases expand in the cylinder.

• Burning gases force the piston to move down. Thus useful work is obtained.

• At the same time, the air in the crank case is compressed by the movement of piston.

• All the ports and the plate valve are in closed position

(d) Exhaust and Transfer:

• At the end of expansion, the exhaust port is uncovered.

• The burnt escape to the atmosphere through the exhaust port.

• Transfer port is also uncovered shortly after the exhaust port is opened.

• The partially compressed air from crank case enters the cylinder the transfer port.

• This air is deflected upwards by the deflected shape of the piston.

• Thus the entering air helps in forcing out the combustion products from the cylinder

• The plate valve remains during this period.

COMPARISON BETWEEN A PETROL ENGINE(SI) AND A DIESEL ENGINE (CI)

S. No.	Petrol engine	Diesel engine
1.	Air-petrol mixture is sucked in the engine cylinder during suction stroke.	Only air is sucked during suction stroke.
2.	Spark plug is used.	Employs an injector.
3.	Power is produced by spark ignition.	Power is produced by compression ignition.
4.	Thermal efficiency up to 25%.	Thermal efficiency up to 40%.

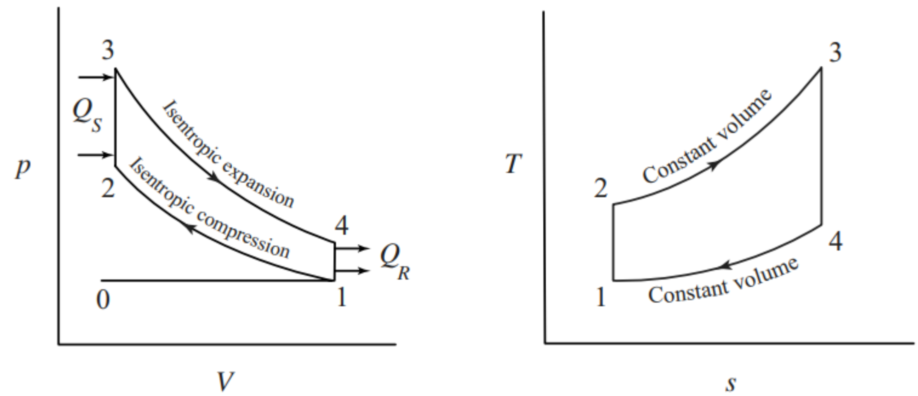
5.	Occupies less space.	Occupies more space.
6.	More running cost.	Less running cost.
7.	Light in weight.	Heavy in weight.
8.	Fuel (Petrol) costlier.	Fuel (Diesel) cheaper.
9.	Petrol being volatile is dangerous.	Diesel is non-dangerous as it is non-volatile.
10.	Pre-ignition possible.	Pre-ignition not possible.
11.	Works on Otto cycle.	Works on Diesel cycle.
12.	Less dependable.	More dependable.
13.	Used in *cars* and *motor cycles*.	Used in heavy duty vehicles like *trucks, buses* and *heavy machinery*.

COMPARISON BETWEEN TWO-STROKE AND FOUR-STROKE ENGINES

Four-Stroke Engine	Two-Stroke Engine
The thermodynamic cycle is completed in four strokes of the piston or in two revolutions of the crankshaft. Thus, one power stroke is obtained in every two revolutions of the crankshaft.	The thermodynamic cycle is completed in two strokes of the piston or in one revolution of the crankshaft. Thus there is one power stroke for every revolution of the crankshaft.
Because of the above, turning moment is not so uniform and hence a heavier flywheel is needed.	Because of the above, turning moment is more uniform and hence a lighter flywheel can be used.
Again, because of one power stroke for two revolutions, power produced for same size of engine is less, or for the same power the engine is heavier and bulkier.	Because of one power stroke for every revolution, power produced for same size of engine is twice, or for the same power the engine is lighter and more compact.
Because of one power stroke in two revolutions lesser cooling and lubrication requirements. Lower rate of wear and tear.	Because of one power stroke in one revolution greater cooling and lubrication requirements. Higher rate of wear and tear.
Four-stroke engines have valves and valve actuating mechanisms for opening and closing of the intake and exhaust valves.	Two-stroke engines have no valves but only ports (some two-stroke engines are fitted with conventional exhaust valve or reed valve).

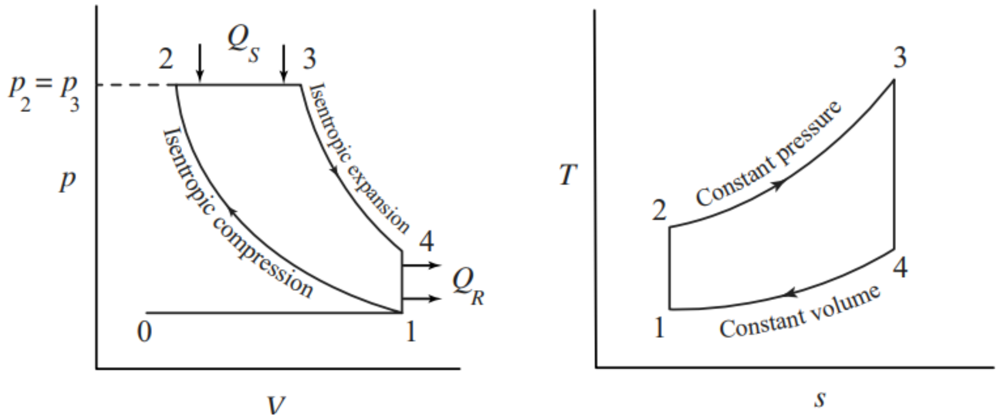
Because of comparatively higher weight and complicated valve mechanism, the initial cost of the engine is more.	Because of light weight and simplicity due to the absence of valve actuating mechanism, initial cost of the engine is less.
Higher volumetric efficiency due to more time for mixture intake.	Lower volumetric efficiency due to lesser time for mixture intake.
Thermal efficiency is higher; part load efficiency is better.	Thermal efficiency is lower; part load efficiency is poor.
Used where efficiency is important, viz., in cars, buses, trucks, tractors, industrial engines, aero planes, power generation etc.	Used where low cost, compactness and light weight are important, viz., in mopeds, scooters, motorcycles, hand sprayers etc.

OTTO CYCLE



- processes 0→1 and 1→0 on the p-V diagram represents suction and exhaust processes and their effect is nullified. The process
- 1→2 represents isentropic compression of the air when the piston moves from bottom dead centre to top dead centre.
- Process 2→3 heat is supplied reversibly at constant volume.
- 3→4 represent isentropic expansion and
- 4→1 and constant volume heat rejection

DIESEL CYCLE:



- 1→2 represents isentropic compression of the air when the piston moves from bottom dead centre to top dead centre.
- Process 2→3 heat is supplied reversibly at constant pressure.
- 3→4 represent isentropic expansion and
- 4→1 and constant volume heat rejection

Refrigeration and Air – conditioning cycles

Refrigeration is the science of producing and maintaining temperatures below that of the surrounding atmosphere.

Important refrigeration applications:

1. Ice making
2. Transportation of foods above and below freezing
3. Industrial air-conditioning
4. Comfort air-conditioning
5. Chemical and related industries
6. Medical and surgical aids
7. Processing food products and beverages
8. Oil refining and synthetic rubber manufacturing
9. Manufacturing and treatment of metals
10. Freezing food products
11. Miscellaneous applications:

(i) Extremely low temperatures

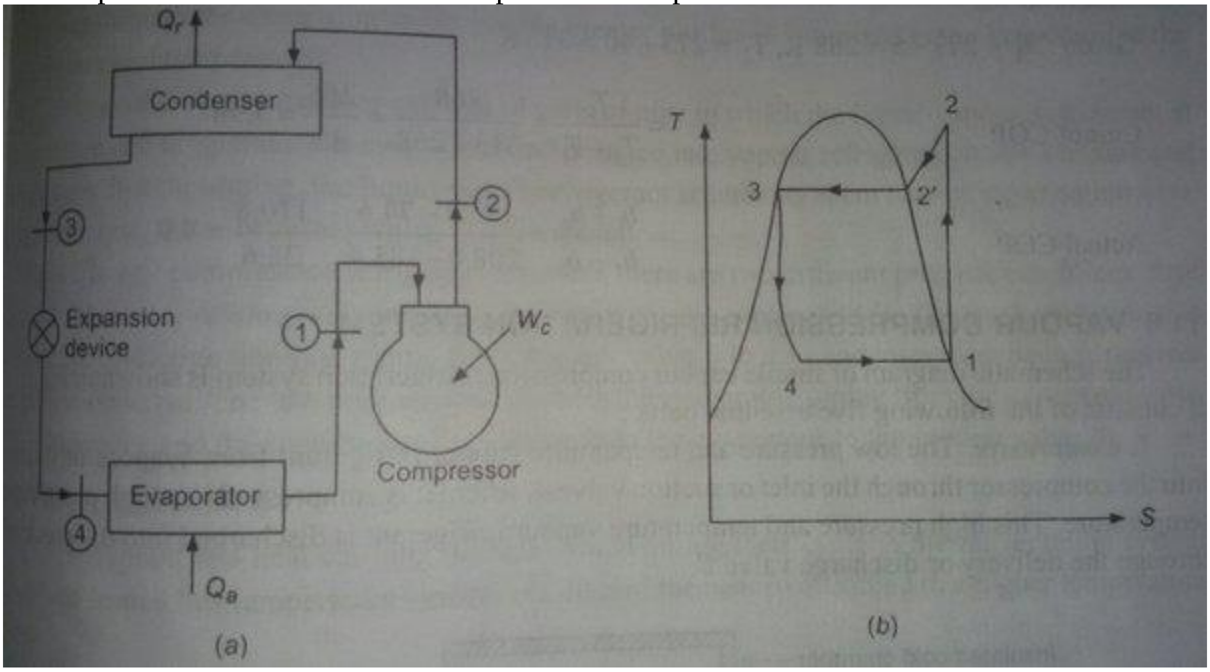
(ii) Plumbing

(iii) Building construction etc.

Tonne of refrigeration (TR): A tone of refrigeration (1TR) is defined as the amount of heat, which is to be extracted from one tonne of water at 0°Cin order to convert into ice at 0°C in 24 hours (day).

$$1 \text{ TR} = 210 \text{ kJ/min} = 3.5 \text{ kW}.$$

Vapour Compression Refrigeration system(VCR): In a simple vapour compression system fundamental processes are completed in :
1. Compression 2. Condensation 3. Expansion 4. Vaporisation**.**



- The vapour at low temperature and pressure enters the "compressor" where it is compressed isentropically and subsequently its temperature and pressure increase considerably.
- This vapour after leaving the compressor enters the ''condenser" where it is condensed into high pressure liquid and is collected in a "receiver tank".
- From receiver tank it passes through the "expansion valve", here it is throttled down to a lower pressure and has a low temperature.
- After finding its way through expansion valve it finally passes on to evaporator where it extracts heat from the surroundings or from the objects to be cooled and vaporises to low pressure vapour.

Merits : 1. C.O.P. is quite high as the working of the cycle is very near to that of reversed Carnot cycle.

2. When used on ground level the running cost of vapour-compression refrigeration system is only 1/5th of air refrigeration system.

3. For the same refrigerating effect the size of the evaporator is smaller.

4. The required temperature of the evaporator can be achieved simply by adjusting the throttle valve of the same unit.

Demerits :1. Initial cost is high.

2. The major disadvantages are in flammability, leakage of vapours and toxicity.

AIR CONDITIONING:

Air Conditioning is the process of conditioning the air according to the human comfort, irrespective of external conditions.

Functions of Air conditioners

- Cleaning air.
- Controlling the temp of air.
- Controlling the moisture content.
- Circulating the air.

Applications of Air Conditioning

- Used in offices, hotels, buses, cars, etc. Used in industries having tool room machines.
- Used in textile industries to control moisture.
- Used in printing press.
- Used in Food industries, Chemical plants.

Classification of air conditioning:

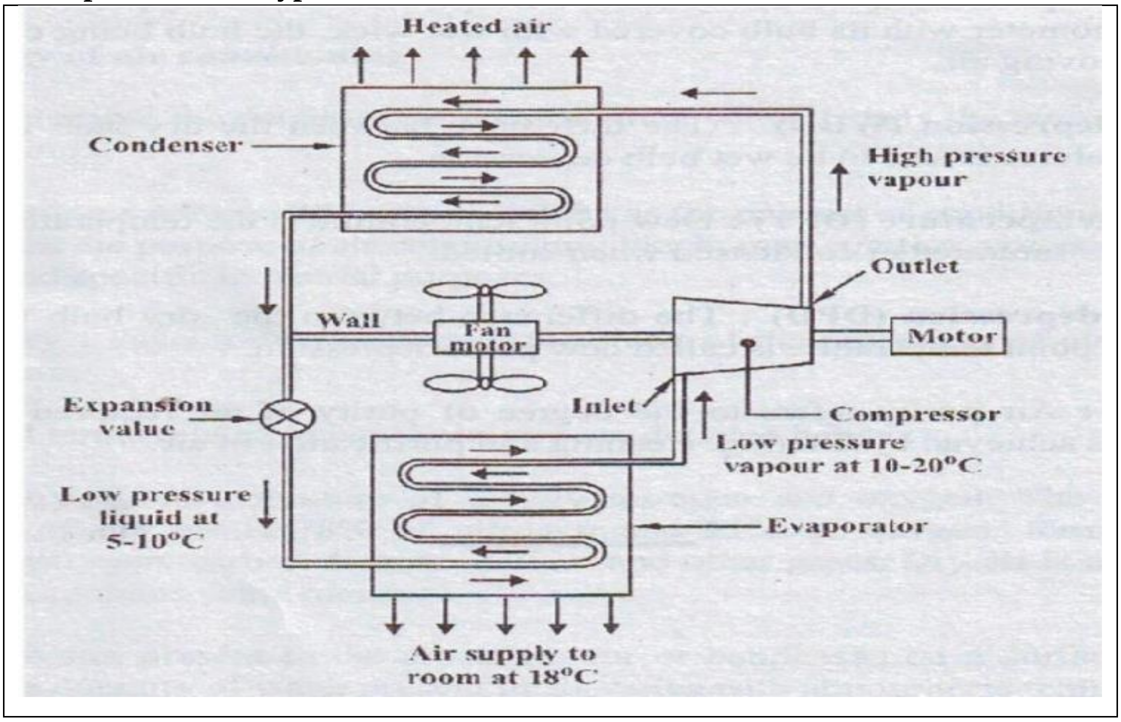
Air conditioning systems are classified as

1) According to the purpose
   - Comfort Air conditioning.
   - Industrial Air conditioning.
2) According to Season of the year
   - Summer Air conditioning.
   - Winter Air conditioning.
   - Year-round Air conditioning.

Types of Air conditioners

- Room Air conditioners
- Winter Air conditioners
- Central Air conditioners

Example: Window type air conditioner



Components of Electric And Hybrid Vehicles.

- Electric vehicles (EVs) are expensive and charging facilities are not fully established. Bridging the gap between conventional and electric cars are the Hybrid Electric Vehicles (HEVs).
- A hybrid vehicle is a type of car that uses two or more different power sources to move the vehicle. The most common type of hybrid vehicle is a gas-electric hybrid, which uses both gasoline and electricity to power the car.
- A hybrid electric vehicle does not plug in to charge. Instead, the battery is charged by the internal combustion engine and via regenerative braking.
- In regenerative braking, the electric motor/generator captures energy normally lost during braking. This recovery and reuse of energy results in better fuel economy without sacrificing performance.
- Energy stored in the battery provides extra power during starts and acceleration. It can also power auxiliary loads and reduce engine idling when the vehicle is stopped

The hybrid car has five main components:

- engine,
- electric motor,
- battery,
- power control unit, and
- regenerative braking system.

Engine: engine is responsible for providing power to the wheels when needed. It typically runs on gasoline or diesel, but some hybrid cars also use alternative fuels such as natural gas or propane.

Electric Motor: The electric motor is used to power the car at lower speeds and during start-up. It is usually powered by a battery, but some hybrid cars also have an onboard generator that can recharge the battery while you are driving.

Battery: The battery is what powers the electric motor and is typically located under the hood of the car. It is important to keep the battery charged in order to maintain optimal performance.

Power Control Unit: The power control unit manages the flow of power between the engine, the electric motor, and the battery. It ensures that the right amount of power is being used at the right time.

Regenerative Braking System: The regenerative braking system captures energy from the brakes and uses it to recharge the battery. This helps to improve fuel efficiency and extends the life of the battery.

Components of Electric Vehicles

1. Traction battery pack
2. DC-DC Converter
3. Electric motor
4. Power inverter
5. Charge Port
6. Onboard charger
7. Controller
8. Auxiliary batteries
9. Thermal system (cooling)
10. Transmission

1. Traction battery pack

A traction battery pack is also known as an Electric vehicle battery (EVB). It powers the electric motors of an electric vehicle. The battery acts as an electrical storage system. It stores energy in the form of DC current. The range will be higher with increasing kW of the battery. The life and operation of the battery depending on its design.

2. DC-DC Converter

The traction battery pack delivers a constant voltage. But different components of the vehicle have different requirements. The DC-DC converter distributes the output power that is coming from the battery to a required level. It also provides the voltage required to charge the auxiliary battery.

3. Electric motor

The electric traction motor is the main component of an electric vehicle. The motor converts electrical energy into kinetic energy. This energy rotates the wheels. An electric motor is the main component that differentiates an electric car from a conventional car. An important feature of an electric motor is the regenerative braking mechanism. This mechanism slows down the vehicle by converting its kinetic energy into another form and storing it for future use. There are basically two types of motors DC and AC motors.

4. Power Inverter

It converts DC power from the batteries to AC power. It also converts the AC current generated during regenerative braking into a DC current. This is further used to recharge the batteries.

5. Charge Port

The charge port connects the electric vehicle to an external supply. It charges the battery pack. The charge port is sometimes located in the front or rear part of the vehicle

6. Onboard charger

The onboard charger is used to convert the AC supply received from the charge port to the DC supply. The onboard charger is located and installed inside the car. It monitors various battery characteristics and controls the current flowing inside the battery pack.

7. Controller

The power electronics controller determines the working of an electric car. It performs the regulation of electrical energy from the batteries to the electric motors. The pedal set by the driver determines the speed of the vehicle and the frequency of variation of voltage that is input to the motor. It also controls the torque produced.

8. Auxiliary batteries

Auxiliary batteries are the source of electrical energy for the accessories in electric vehicles. Without the main battery, the auxiliary batteries will continue to charge the car. It prevents the voltage drop, produced during engine start from affecting the electrical system.

9. Thermal system(cooling)

The thermal management system is responsible for maintaining an operating temperature for the main components of an electric vehicle such as an electric motor, controller, etc. It functions during charging as well to obtain maximum performance. It uses a combination of thermoelectric cooling, forced air cooling, and liquid cooling.

10. Transmission

It is used to transfer the mechanical power from the electric motor to the wheels, through a gearbox. The advantage of electric cars is that they do not require multi-speed transmissions. The transmission efficiency should be high to avoid power loss.